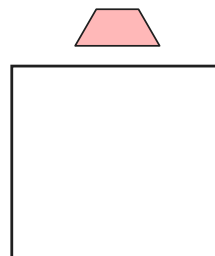
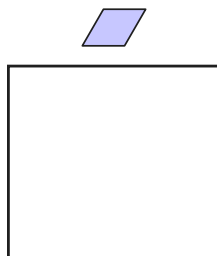
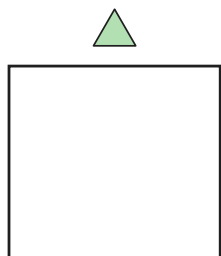
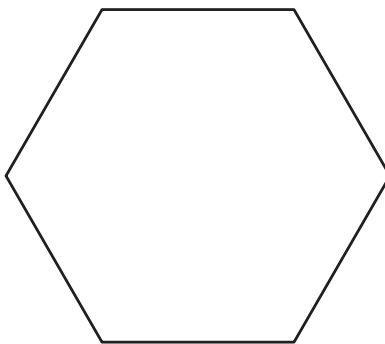
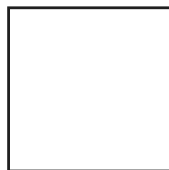
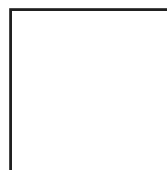
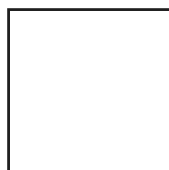
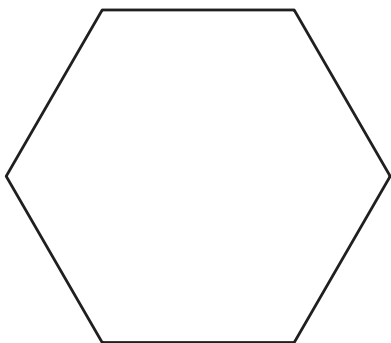
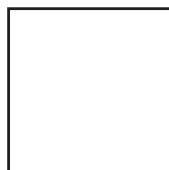
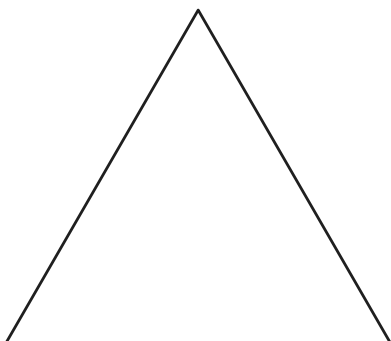
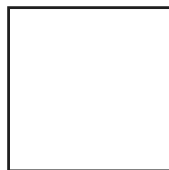
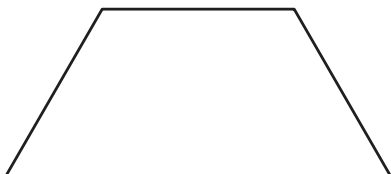


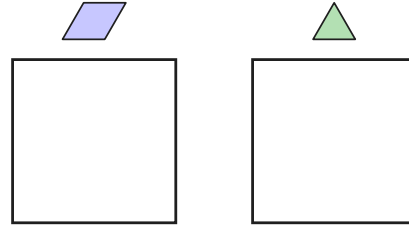
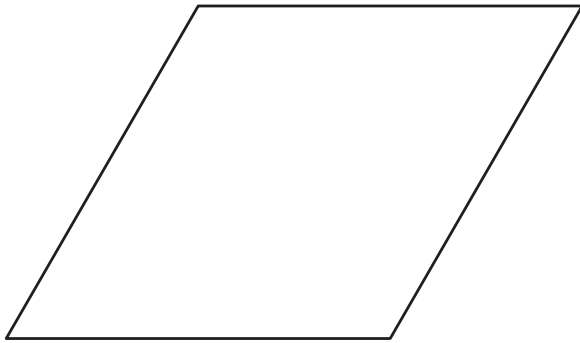
Problem 1: Cover the hexagon with green triangles . Do not leave gaps. Do not let pieces overlap or stick out past the line. Count the green triangles you used and write the number in the first box. Then cover the same hexagon with blue rhombi  only, and write how many in the second box. Then cover it with red trapezoids  only, and write how many in the third box.



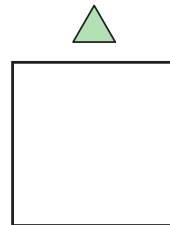
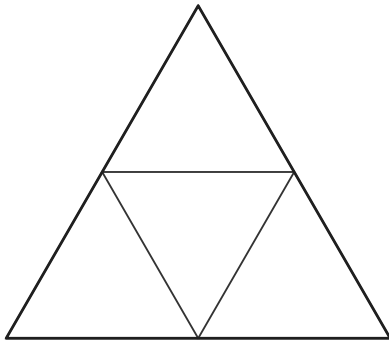
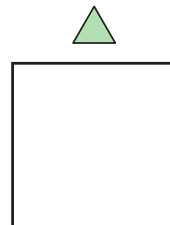
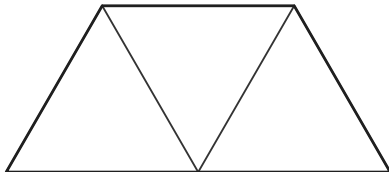
Problem 2: Cover shape A with blue rhombi  only. If you can cover it, draw a check mark in the box under the blue rhombus. If you cannot, draw an X there. Then cover shape A with green triangles , count them, and write the number in the box under the green triangle. Do the same for shapes B, C and D.

A**B****C****D**

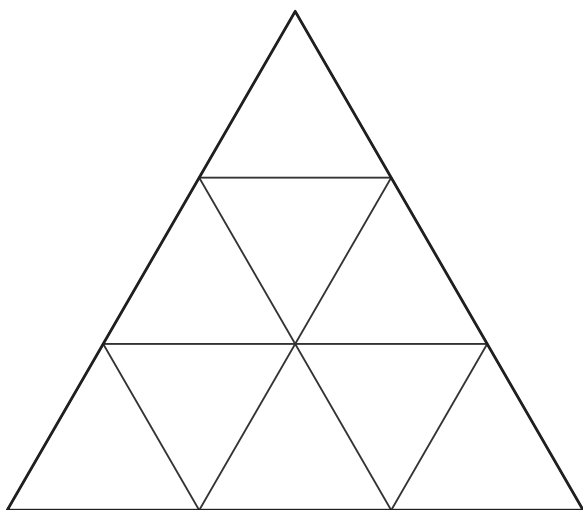
Problem 3: Do the same for shape E: blue rhombi  only, then a check mark or an X in the first box; then green triangles , and the number in the second box.

E

Problem 4: Cover shape C with four green triangles , one in each little triangle. Now trade: take off two green triangles that touch along a whole side, and put one blue rhombus  in their place. Keep trading until no two green triangles touch any more. Count the green triangles that are left and write the number in the box. Then do the same on shape D.

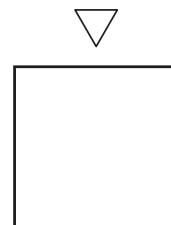
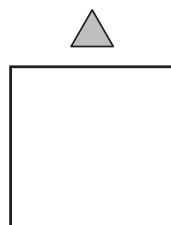
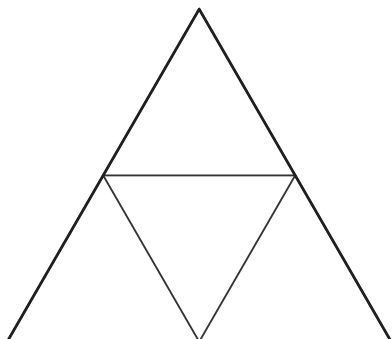
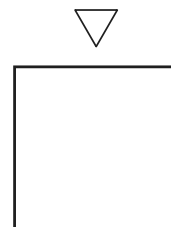
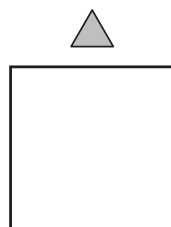
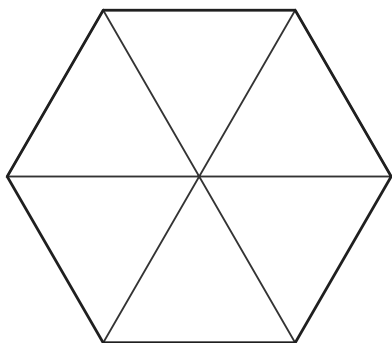
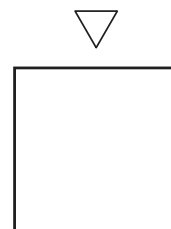
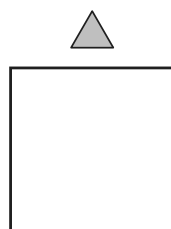
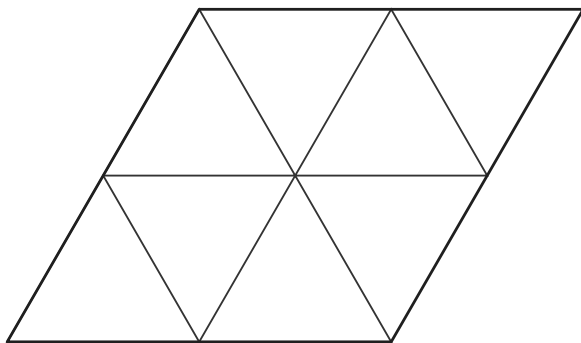
C**D**

Problem 5: Shape G is a big triangle cut into little triangles. Color every little triangle that points up, like $\triangle$. Leave every little triangle that points down, like ∇ , white. Then put one blue rhombus $\textcolor{blue}{\parallel\!\!\!/\!}$ anywhere on shape G so that it covers two little triangles exactly. Count how many colored and how many white triangles are under it, and write the two numbers in row 1 of the table. Turn the rhombus and put it somewhere else. Count again and fill row 2. Do this five times.

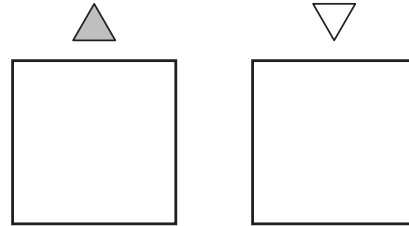
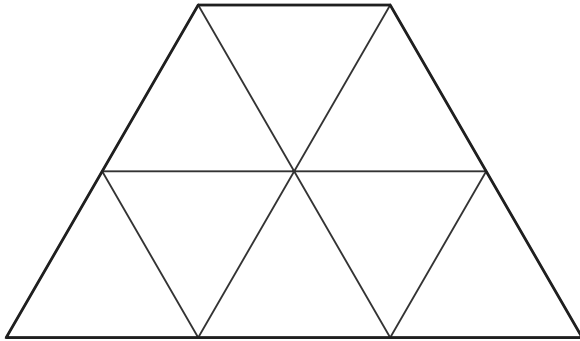
G

		
1		
2		
3		
4		
5		

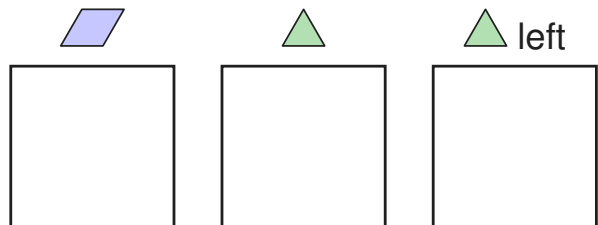
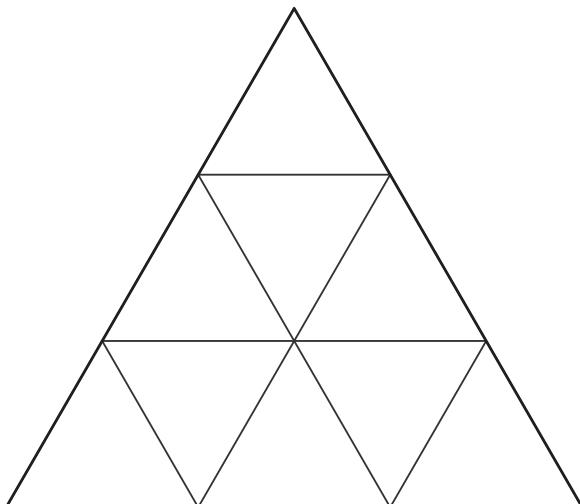
Problem 6: Color every up-pointing triangle $\triangle$ in shapes C, B and E. Leave the down-pointing ones ∇ white. Count the colored triangles and the white triangles in each shape and write the two numbers in its boxes.

C**B****E**

Problem 7: Shape F is new. First color its up-pointing triangles  and write the two numbers in the first two boxes. Then, before you touch any pieces, draw a check mark or an X in the box that says *guess*: a check mark if you think four blue rhombi  can cover shape F, an X if you think they cannot. Then try it with the pieces and draw a check mark or an X in the box that says *try*.

F*guess**try*

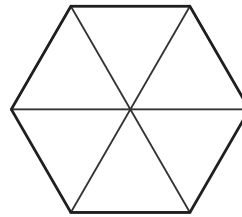
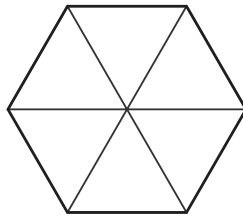
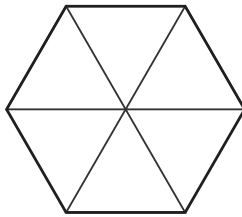
Problem 8: Cover shape G with blue rhombi  only, and draw a check mark or an X in the first box. Then cover shape G with green triangles , count them, and write the number in the second box. Now trade as in Problem 4: take off two green triangles that touch along a whole side and put one blue rhombus in their place, until no two green triangles touch. Write the number of green triangles that are left in the third box.

G

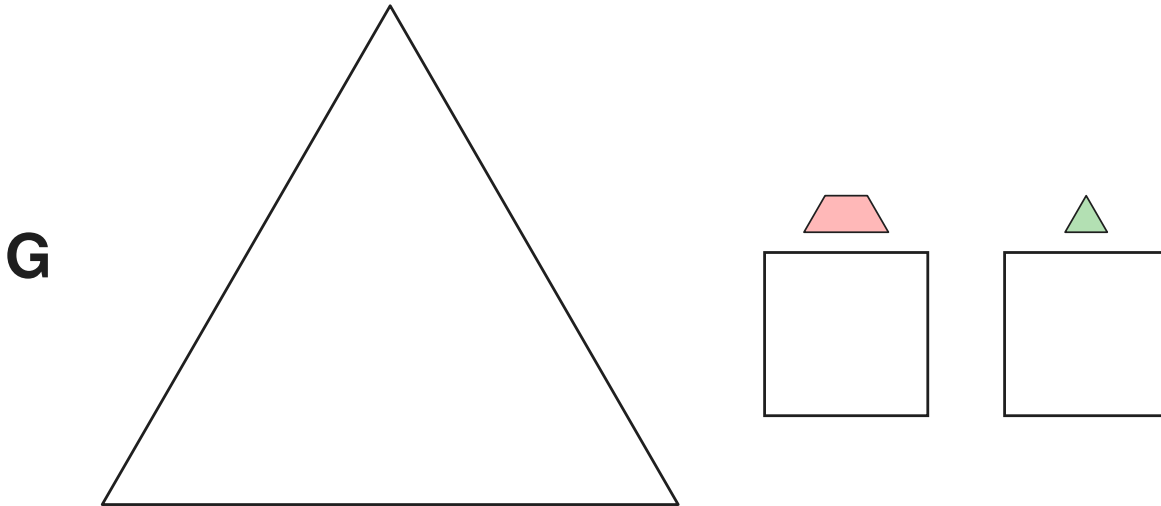
Problem 9: Write the number of green triangles you wrote for each shape in Problems 2, 3 and 8 in the box next to its letter. For each shape, take that many green triangles and put them in pairs, two and two. Circle the letter of every shape that has one triangle left over with no partner.

A		B		C		D		E		G	
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Problem 10: Cover shape B, the hexagon on page 2, with three blue rhombi . Draw your three rhombi on the first small hexagon: draw a dark line along every edge where two of your rhombi meet. Now cover the hexagon in a different way and draw it on the second small hexagon. Try to find a third way and draw it on the third hexagon.



Problem 11: Cover shape G, the big triangle, with red trapezoids  only. Count them and write the number in the first box. Then cover shape G again, this time using blue rhombi  and green triangles  together. Use as few green triangles as you can. Write the smallest number of green triangles you managed in the second box.



Problem 12: Shape I is a bigger triangle. Cover it with any pieces you like. Count the pieces. Then cover it again with fewer pieces. Write the smallest number of pieces you managed in the box.

